

WIPRO CAPSTONE PROJECT

Test Analysis Report

BlazeDemo Flight Booking Automation Framework

Selenium WebDriver · TestNG · Maven · ExtentReports · Jenkins · Docker

Project Name	BlazeDemo Flight Booking Automation Framework
Document Type	Test Analysis Report
Prepared By	Mr. Baivab Sarkar
Mentor/ Trainer	Mrs. Vaishali Sonanis
Application Under Test	BlazeDemo
URL	https://blazedemo.com/
Positive Suite Result	PASS: 12/12 (100%)
Negative Suite Result	PASS: 3/3 (100%)
Defects Logged	7 (BLAZE-7 to BLAZE-13)
Repository	ThisIs-Developer/Wipro-Capstone-Project
GitHub Link	https://github.com/ThisIs-Developer/Wipro-Capstone-Project
Date	13 th June 2026

1. Executive Summary

This Test Analysis Report summarizes the results of testing conducted on the BlazeDemo Flight Booking web application. Testing was performed using Selenium WebDriver with the Page Object Model (POM) framework and TestNG. Two test suites were executed: a positive suite that tested the complete flight booking process with three sets of passenger data, and a negative test suite that checked the system's response to invalid inputs and Authentication module.

All test cases passed successfully, showing that the main booking process works as expected. However, 7 defects were identified and logged in JIRA.

2. Scope of Testing

The following modules were included in the testing scope:

- Home Page
- Flight Search
- Flight Selection
- Purchase Page
- Booking Confirmation
- Registration Page
- Login Page

The testing focused on validating critical business workflows and identifying application defects.

3. Testing Types Performed

- **Functional Testing:** Validated application functionality against expected behavior.
- **End-to-End Testing:** Verified the complete flight booking workflow from flight search to booking confirmation.
- **Data-Driven Testing:** Executed booking scenarios using multiple passenger datasets from Excel.
- **Negative Testing:** Validated application behavior using invalid, blank, and malformed inputs.
- **Smoke Testing:** Verified critical application functionalities before detailed execution.
- **Regression Testing:** Ensured framework stability across multiple executions.
- **Cross-Browser Testing:** Verified application functionality across different web browsers to ensure consistent behavior.

4. Test Case Design

Test Case ID	Test Case Name	Module	Status
TC001	Verify Home Page	Home Page Module	PASS
TC002	Verify Flight Search	Flight Search Module	PASS
TC003	Verify Purchase Page	Purchase Module	PASS
TC004	Verify Booking Confirmation	Confirmation Module	PASS
TC005	Verify Registration Page	Authentication Module	FAIL
TC006	Verify Login Page	Authentication Module	FAIL

Test Case ID	Test Case Name	Module	Status
TC007	Booking With Blank Mandatory Fields	Purchase Module	FAIL
TC008	Invalid Passenger Information	Purchase Module	FAIL
TC009	Invalid Month Validation	Payment Module	FAIL
TC010	Expired Year Validation	Payment Module	FAIL
TC011	Mandatory Field Indicator	Purchase Module	FAIL
TC012	Credit Card Length Validation	Payment Module	FAIL

5. Test Case Execution Analysis

a) Positive Test Cases

TC ID	Test Case	Expected Result	Actual Result	Status
TC001	Verify Home Page	Home page loads successfully	Home page loaded successfully	PASS
TC002	Verify Flight Search	Available flights displayed	Flights displayed successfully	PASS
TC003	Verify Purchase Page	Form submitted successfully	Form submitted successfully	PASS
TC004	Verify Booking Confirmation	Confirmation page displayed	Confirmation page displayed	PASS

b) Negative Test Cases

TC ID	Test Case	Expected Result	Actual Result	Status	Bug No (JIRA)
TC005	Verify Registration Page	User registration successful	419 Page Expired	FAIL	BLAZE-13
TC006	Verify Login Page	User login successful	419 Page Expired	FAIL	BLAZE-13
TC007	Blank Mandatory Fields	Validation message displayed	Booking confirmed	FAIL	BLAZE-7
TC008	Invalid Passenger Information	Validation message displayed	Booking confirmed	FAIL	BLAZE-8
TC009	Invalid Month Validation	Month rejected	Booking confirmed	FAIL	BLAZE-9
TC010	Expired Year Validation	Year rejected	Booking confirmed	FAIL	BLAZE-10
TC012	Credit Card Length Validation	Validation message displayed	Booking confirmed	FAIL	BLAZE-12

Note: Although these test cases are marked as failed from an application perspective, they are considered successful from a testing perspective because they successfully identified reproducible defects.

c) UI Test Cases

TC ID	Test Case	Expected Result	Actual Result	Status	Defect
TC011	Mandatory Field Indicator	Mandatory fields marked	No indication available	FAIL	BLAZE-11

6. Suite-Level Test Execution Results

Test Suite	Suite File	Tests	Passed	Failed	Skipped	Result
FlightBookingSuite	testng.xml	12	12	0	0	PASS
NegativeTestSuite	testng-negative.xml	3	3	0	0	PASS
TOTAL	NA	15	15	0	0	PASS

7. Defects Logged in JIRA

 BLAZE-7	Booking can be completed with fields left blank	BLAZEDEMO FLIGHT ...	TO DO ▾	Jun 15	   
 BLAZE-8	Purchase Flight Form Does Not Perform Input Validation on Passenger and Payment ...	BLAZEDEMO FLIGHT ...	TO DO ▾	Jun 15	   
 BLAZE-9	Month field accepts invalid values outside valid range	BLAZEDEMO FLIGHT ...	TO DO ▾	Jun 15	  
 BLAZE-10	Expired year (1900) displayed as default credit card expiry year	BLAZEDEMO FLIGHT ...	TO DO ▾	Jun 15	  
 BLAZE-11	Mandatory fields are not visually identified	BLAZEDEMO FLIGHT ...	TO DO ▾	Jun 15	  
 BLAZE-12	Credit Card Number field does not enforce minimum and maximum length validation	BLAZEDEMO FLIGHT ...	TO DO ▾	Jun 15	  
 BLAZE-13	Login and Registration Pages Return "419 Page Expired" Error	BLAZEDEMO FLIGHT ...	TO DO ▾	Jun 15	  

8. JIRA Defect Report

Defect ID	Title	Severity	Priority	Module	Status
BLAZE-7	Booking can be completed with fields left blank	High	High	Purchase Page	TO DO
BLAZE-8	Purchase Flight Form Does Not Perform Input Validation on Passenger and Payment Information	High	High	Purchase Page	TO DO
BLAZE-9	Month field accepts invalid values outside valid range	Medium	Medium	Purchase Page	TO DO
BLAZE-10	Expired year (1900) displayed as default credit card expiry year	Medium	Medium	Purchase Page	TO DO
BLAZE-11	Mandatory fields are not visually identified	Low	Low	Purchase Page	TO DO
BLAZE-12	Credit Card Number field does not enforce minimum and maximum length validation	Medium	High	Purchase Page	TO DO
BLAZE-13	Login and Registration Pages Return "419 Page Expired" Error	High	High	Login Page / Registration Page	TO DO

9. Test Output

Run as TestNG Suite in IDE

The screenshot displays the Eclipse IDE interface during a TestNG test run. The left pane shows the 'Console' window with the following output:

```
terminated> BlazeDemo_testng.xml [TestNG] C:\Users\User\p2\pool\plugins\org.eclipse.justj.openj
Jun 12, 2026 12:18:28 PM org.openqa.selenium.devtools.CdpVersionFinder fin
WARNING: Unable to find an exact match for CDP version 149, returning the
2026-06-12 12:18:28 [main] INFO Log - Launching Browser : chrome
2026-06-12 12:18:30 [main] INFO Log - Application Opened
STARTED : verifyHomePage
2026-06-12 12:18:30 [main] INFO Log - Home Page Verification Started
2026-06-12 12:18:31 [main] INFO Log - Home Page Verification Completed
PASSED : verifyHomePage
STARTED : verifyFlightSearch
2026-06-12 12:18:31 [main] INFO Log - Flight Search Verification Started
2026-06-12 12:18:32 [main] INFO Log - Flight Search Verification Complete
PASSED : verifyFlightSearch
STARTED : verifyPurchasePage
2026-06-12 12:18:32 [main] INFO Log - Purchase Page Verification Started
2026-06-12 12:18:34 [main] INFO Log - Purchase Page Verification Complete
PASSED : verifyPurchasePage
STARTED : verifyBookingConfirmation
2026-06-12 12:18:34 [main] INFO Log - Confirmation Verification Started
2026-06-12 12:18:34 [main] INFO Log - Screenshot Captured
2026-06-12 12:18:34 [main] INFO Log - Booking Confirmation Completed
PASSED : verifyBookingConfirmation
2026-06-12 12:18:34 [main] INFO Log - Closing Browser

=====
lightBookingSuite
Total tests run: 12, Passes: 12, Failures: 0, Skips: 0
=====
```

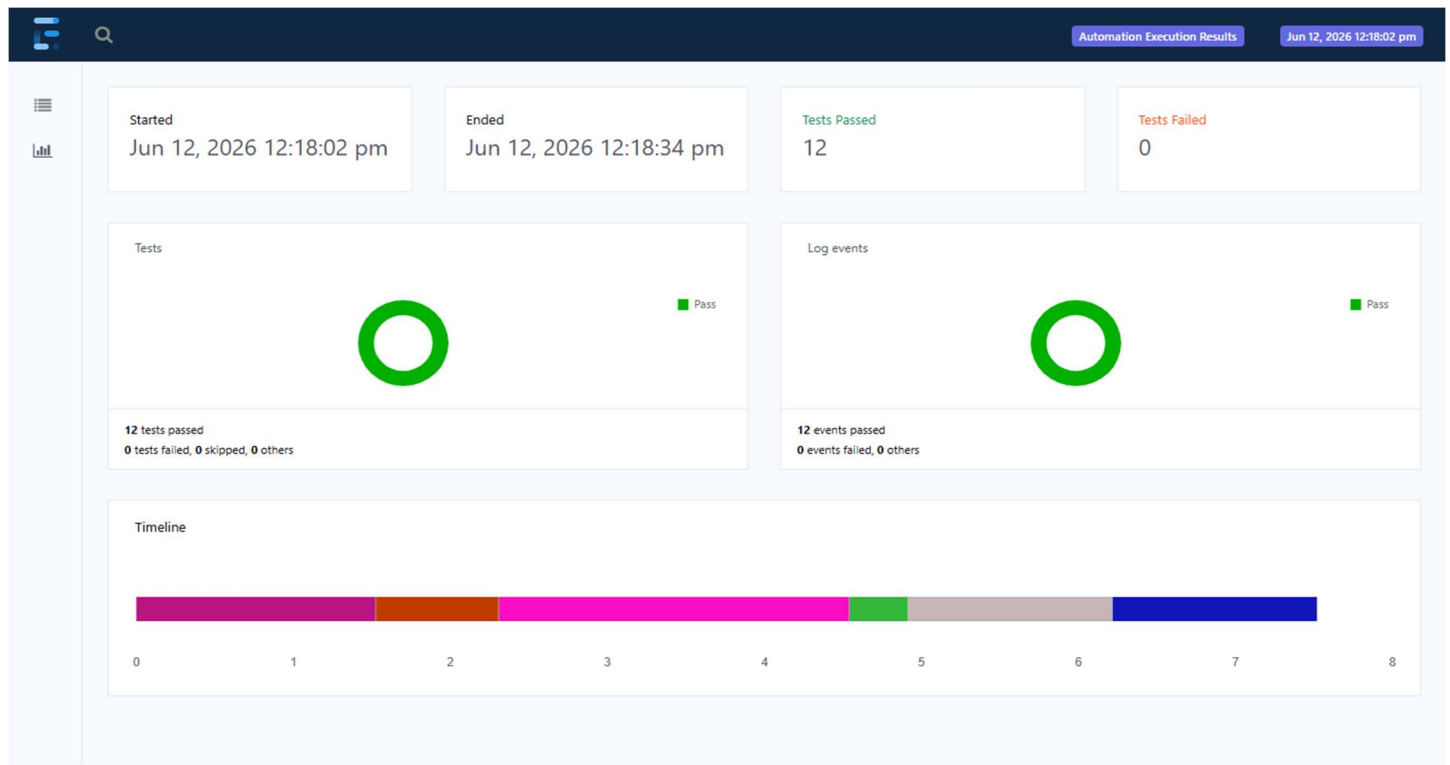
The right pane shows the 'Results of running suite' window. It displays a summary of the test results:

- Search:
- Passed: 12
- Failed: 0
- Skipped: 0
- 1/1 Methods: 12 (37997 ms)

The 'All Tests' tab is selected, showing a tree view of the test suite:

- FlightBookingSuite (12/0/0/0) (15.332 s)
 - FlightBookingTests (15.332 s)
 - testcases.FlightBookingTest
 - verifyHomePage (1.563 s)
 - verifyFlightSearch (0.409 s)
 - verifyPurchasePage (2.825 s)
 - verifyBookingConfirmation (1.059 s)
 - verifyHomePage (1.308 s)
 - verifyFlightSearch (0.779 s)
 - verifyPurchasePage (2.172 s)
 - verifyBookingConfirmation (0.428 s)
 - verifyHomePage (1.313 s)
 - verifyFlightSearch (0.801 s)
 - verifyPurchasePage (2.239 s)
 - verifyBookingConfirmation (0.436 s)

ExtentReport (.html)



Mvn clean test

```
2026-06-13 10:37:43 [main] INFO Log - Home Page Verification Completed
PASSED : verifyHomePage
STARTED : verifyFlightSearch
2026-06-13 10:37:43 [main] INFO Log - Flight Search Verification Started
2026-06-13 10:37:44 [main] INFO Log - Flight Search Verification Completed
PASSED : verifyFlightSearch
STARTED : verifyPurchasePage
2026-06-13 10:37:44 [main] INFO Log - Purchase Page Verification Started
2026-06-13 10:37:47 [main] INFO Log - Purchase Page Verification Completed
PASSED : verifyPurchasePage
STARTED : verifyBookingConfirmation
2026-06-13 10:37:47 [main] INFO Log - Confirmation Verification Started
2026-06-13 10:37:48 [main] INFO Log - Screenshot Captured
2026-06-13 10:37:48 [main] INFO Log - Booking Confirmation Completed
PASSED : verifyBookingConfirmation
2026-06-13 10:37:48 [main] INFO Log - Closing Browser
[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 52.65 s -- in TestSuite
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 01:03 min
[INFO] Finished at: 2026-06-13T10:37:49+05:30
[INFO] -----
```

Jenkins runs

 **Jenkins** / Project BlazeDemo / #5 / Stages

 < #5 Rerun 

Started by Baivab Sarkar Started 10 hr ago Queued 43 ms Took 3 min 45 sec

Start ○ — ○ Checkout SCM — ○ Tool Install — ○ Checkout — ○ Build — ○ Compile — ○ Test — ○ Docker Build — ○ Docker Run — ○ Post Actions — ○ End

Search

- Checkout SCM 2s
- Tool Install 0.36s
- Checkout 3s
- Build 10s
- Compile 7s
- Test 45s
- Docker Build 1m 37s
- Docker Run 46s
- Post Actions 0.14s

 **Post Actions** 0.14s Started 10h ago Jenkins

-  Pipeline Execution Completed 48ms
 - 0 Pipeline Execution Completed
-  Build Successful 34ms
 - 0 Build Successful

Docker Image Run and Logs

Gordon

Containers

Images

Logs

Volumes

Kubernetes

Builds

Models

MCP Toolkit

Docker Hub

Extensions

Upgrade plan

Containers / stoic_mclean

stoic_mclean

a84d1e26f28d

blazedemo-framework:latest

STATUS

Exited (0) (4 minutes ago)

Logs

Inspect

Bind mounts

Exec

Files

Stats

View and search logs for all containers in one place with the new Logs view.

View logs

PASSED : verifyPurchasePage

STARTED : verifyBookingConfirmation

2026-06-12 07:00:32 [main] INFO Log - Confirmation Verification Started

2026-06-12 07:00:32 [main] INFO Log - Screenshot Captured

2026-06-12 07:00:32 [main] INFO Log - Booking Confirmation Completed

PASSED : verifyBookingConfirmation

2026-06-12 07:00:32 [main] INFO Log - Closing Browser

[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 39.87 s -- in TestSuite

[INFO]

[INFO] Results:

[INFO]

[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0

[INFO]

[INFO] -----

[INFO] BUILD SUCCESS

[INFO] -----

[INFO] Total time: 48.669 s

[INFO] Finished at: 2026-06-12T07:00:33Z

[INFO] -----

Engine running

RAM 1.80 GB CPU 0.00% Disk: 15.15 GB used (limit 1006.85 GB)

Terminal v4.77.0

Logs

Give feedback

Add search filter (supports regex like */error/warn/*)

Aa

+

Timestamp	Object	Message
2026-06-12 12:30:32	stoic_mclean	2026-06-12 07:00:32 [main] INFO Log - Screenshot Captured
2026-06-12 12:30:32	stoic_mclean	2026-06-12 07:00:32 [main] INFO Log - Booking Confirmation Completed
2026-06-12 12:30:33	stoic_mclean	PASSED : verifyBookingConfirmation
2026-06-12 12:30:33	stoic_mclean	2026-06-12 07:00:32 [main] INFO Log - Closing Browser
2026-06-12 12:30:33	stoic_mclean	[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: ...
2026-06-12 12:30:33	stoic_mclean	[INFO]
2026-06-12 12:30:33	stoic_mclean	[INFO] Results:
2026-06-12 12:30:33	stoic_mclean	[INFO]
2026-06-12 12:30:33	stoic_mclean	[INFO] Tests run: 12, Failures: 0, Errors: 0, Skipped: 0
2026-06-12 12:30:33	stoic_mclean	[INFO]
2026-06-12 12:30:33	stoic_mclean	[INFO] -----
2026-06-12 12:30:33	stoic_mclean	[INFO] BUILD SUCCESS
2026-06-12 12:30:33	stoic_mclean	[INFO] -----
2026-06-12 12:30:33	stoic_mclean	[INFO] Total time: 48.669 s
2026-06-12 12:30:33	stoic_mclean	[INFO] Finished at: 2026-06-12T07:00:33Z
2026-06-12 12:30:33	stoic_mclean	[INFO] -----

9882 lines

Stick to bottom

Engine running

RAM 1.82 GB CPU 0.50% Disk: 15.15 GB used (limit 1006.85 GB)

Terminal v4.77.0

5. Conclusion

The BlazeDemo Flight Booking Automation Framework successfully validated the application's core booking functionality and achieved 100% test execution success. Testing identified 7 defects related to authentication and validation, while confirming that the end-to-end booking workflow functions correctly. Overall, the project met its testing objectives and demonstrated an effective, scalable automation framework with Jenkins and Docker integration.

Page 7 of 7